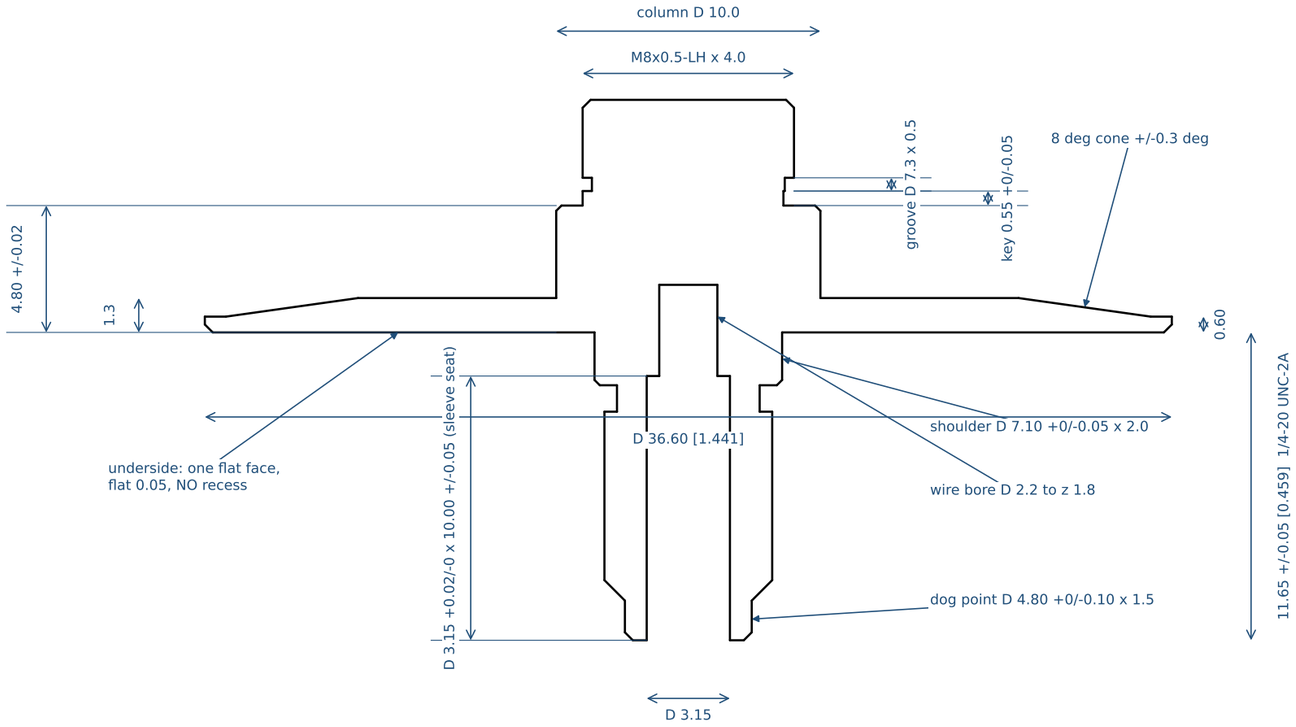
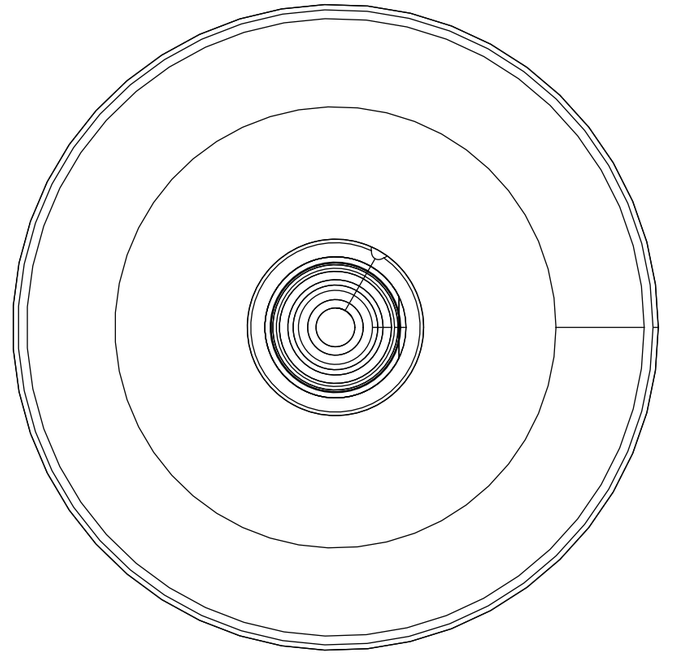
SECTION THROUGH THE AXIS (y=0); z = 0 at the contact (underside) face



TOP VIEW (spigot with D-flat, column, cone, window seat)



NOTES:

1. Stud 1/4-20 UNC-2A, length 11.65 +/-0.05 from the contact (underside) face. TIP: dog point D 4.80 +0/-0.10 x 1.5, 0.3 x 45 chamfer. Smooth shoulder D 7.10 +0/-0.05 x 2.0 under the flange (the record centres on it), relief D 5.4 x 1.0 below it.
2. UNDERSIDE IS ONE FLAT FACE, D 36.6, flat within 0.05, square to the stud within 0.05: NO felt recess (rev A's D 13.2 x 1.0 recess is deleted). It sits on a PET washer.
3. STEPPED BORE, concentric to the thread within 0.05: D 3.15 +0.02/-0 x 10.00 +/-0.05 deep from the tip (press fit for the 3.175 PEEK sleeve; the square shoulder is the sleeve's STOP), then D 2.2 to z 1.8. Deburr the bore mouth; nothing may stand proud of the tip face. Do NOT stake.
4. Column top at 4.80 +/-0.02 above the contact face, flat within 0.03 (LED PCB seat + return contact; lip-gap datum).
5. Spigot M8x0.5 LEFT-HAND, 4.0 long: first 0.55 +0/-0.05 is an UNTHREADED D 8.0 with a flat at 3.6 from the axis (keys the LED PCB; must stay BELOW the 0.72 min PCB), then a run-out groove D 7.3 x 0.5, then thread.
6. Wire cross-hole D 1.0 at 45 deg, az 60: axis passes (r 5.0, z 4.6), so it breaks out through the EDGE of the column top face (r 4.5..5) and meets the D 2.2 bore near z 0.7. Deburr both ends flush: a burr here sits on the PCB seat.
7. Louver floor: 8 deg cone from (r 12.5, z 1.3) to (r 17.5, z 0.60), +/-0.3 deg. Loctite 270 on the spigot needs activator 7649 on nickel; keep it off the column top.

RECORD SCONCE | LIT CAP CORE: hollow 1/4-20 stud + flange + column

PART NO: RS-CAP-12L-CORE REV: B DATE: 2026-11-11 SUPERSEDES: rev A, A.1 (do not build or quote)

MATERIAL: Brass C360 (CDA 360) bar, turned + one 45 deg cross-hole (live tooling or 2nd op).

FINISH: Electroless nickel 5 um MAX, EVERYWHERE (conductive; thread class is before plating, it must accept a 1/4-20 2B GO gauge after). NO chrome. Then PAINT ONLY, masking the thread, shoulder, bore, underside (contact) face, column top and spigot: white reflector paint (R >= 0.90) on the flange top r 5..14 and the column side; matte black on the cone r 14..17.5, window seat and rim OD.

UNITS: mm [in] THIRD-ANGLE PROJECTION SCALE: as shown

GENERAL TOLERANCES: x.x +/-0.3 x.xx +/-0.1 angles +/-0.5 deg unless noted